Research Article

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Empirical model integration for accurate charge carrier mobility simulation in silicon MOSFETs

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Abstract: This study investigates the use of combined empirical models to improve charge carrier mobility simulation in silicon MOSFETs, overcoming the limitations of conventional single-model approaches. With continued device miniaturization, accurate mobility modeling is essential to capture interactions among temperature, doping concentration, electric fields, and scattering mechanisms. Three hybrid modeling strategies are assessed, (i) Power Law–Lombardi, (ii) Arora–Lombardi–Caughey-Thomas, and (iii) Arora–Fletcher–Lombardi–Caughey-Thomas, against a baseline power-law model. Numerical results indicate that the Arora–Lombardi–Caughey-Thomas combination achieves the best balance between accuracy and computational efficiency, effectively representing doping, temperature, and high-field effects. Validation with experimental data confirms the model's predictive reliability. The findings emphasize the advantage of integrating empirical models to enhance both precision and performance in MOSFET simulations.

Keywords: charge carrier mobility; silicon MOSFETs; empirical models; model integration; semiconductors

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1 Introduction

Continuous progress in semiconductor device technology has driven the scaling of metal–oxide–semiconductor field-effect transistors (MOSFETs) into nanometer regimes, where classical models fail to capture the increasingly complex physical behaviors. Among emerging power semiconductor technologies, silicon carbide (SiC) and gallium nitride (GaN) have demonstrated outstanding capabilities in high-frequency, high-voltage, and high-temperature environments owing to their wide bandgap, high critical electric field, and excellent thermal conductivity [1–4]. Bonkoungou et al. [1] developed a semi-physical EKV model for SiC MOSFETs capable of describing nonlinear switching behavior, while Shankari et al. [2] examined vertical GaN MOSFET architectures to enhance device robustness under elevated thermal stress. Dai et al. [3] provided transient electrothermal insights into SiC devices, and Er-rafii and Galadi [4] formulated a temperature-scalable DC model that accurately predicts electrical characteristics over broad operational ranges.

The necessity for high-reliability and radiation-tolerant power devices has also intensified. Peng et al. [5] explored the impact of total ionizing dose (TID) effects in poly-Si thin-film transistors (TFTs), whereas Shen and Yuan [6] applied Green's function-based methods to investigate low-frequency noise mechanisms in MOSFETs. In the context of nanoscale biosensors, Ding et al. [7] introduced a junctionless cylindrical surrounding-gate (JLCSG) MOSFET integrating quantum confinement effects for enhanced sensing precision. Cabré et al. [8] developed a semi-empirical predictive model for double-gate MOSFETs that effectively couples electrostatics and carrier transport, and Wong [9] presented a universal framework for carrier mobility

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modeling, emphasizing its dependency on vertical electric fields.

Quantum mechanical effects have been increasingly incorporated into analytical MOSFET formulations to enhance physical accuracy. Peng et al. [10] developed a compact model for nanoscale gate-all-around MOSFETs that explicitly accounts for quantum confinement, while Ortiz-Conde et al. [11] derived a rigorous surface potential equation based on Fermi–Dirac statistics using polylogarithmic functions, thereby improving predictive precision for deep submicron devices. Similarly, Ghosh [12] proposed an analytical framework for short-channel, fully depleted silicon-on-insulator (SOI) MOSFETs, revealing how geometry scaling influences inversion charge and electrostatic control. Alessandrini et al. [13] extended this concept by developing a comprehensive mobility model for both single- and double-gate SOI MOSFETs, validated against experimental data. Ye et al. [14] conducted a two-dimensional simulation study on silicon power MOSFETs operating at low temperatures, highlighting the importance of temperature-dependent transport models for accurately describing carrier dynamics and device performance. Balaguer et al. [15] modeled inversion charge in double-gate MOSFETs, while Baral et al. [16] introduced a unified two-dimensional model for nanowire junctionless MOSFETs with strong electrostatic coupling.

Moldovan et al. [17] presented compact charge and capacitance models applicable to modern transistor geometries, and Roldán and Gámiz [18] analyzed strained-Si/SiGe structures to understand mobility enhancement mechanisms. In parallel, Toan and Maity [19] developed physics-based models for III–V double-gate MOSFETs, while Palestri et al. [20] applied semi-classical Monte Carlo simulations to study transport behavior under nanoscale confinement. Madheswaran et al. [21] investigated hot-carrier reliability and optical properties of advanced MOSFET structures, and He et al. [22] presented the BSIM5 model, an advanced charge-based MOSFET model developed for accurate nanoscale VLSI circuit simulation.

Further developments have focused on channel material engineering and quantum confinement. Bhattacherjee and Biswas [23] analyzed long-channel germanium double-gate *p*-MOSFETs with high-*k* dielectrics, showing that dielectric–semiconductor interface quality and carrier effective mass significantly affect subthreshold behavior. Osgnach et al. [24] examined the impact of interface states on electron mobility in InGaAs MOSFETs, while Cerdeira et al. [25] proposed a compact model for doped short-channel double-gate devices. Wang et al. [26] developed a compact model to account for quantum effects in symmetric double-gate MOSFETs, while Fei et al. [27] investigated the temperature limitations and failure mechanisms of SiC MOSFETs under avalanche conditions, emphasizing the thermal–electrical trade-offs at high bias operation.

Venugopalan et al. [28] developed the BSIM-CG compact model for cylindrical and surround-gate MOSFETs, incorporating a surface potential-based formulation with quantum confinement effects. Ratnesh et al. [29] reviewed the scaling challenges in MOSFET technologies, emphasizing performance degradation caused by short-channel, tunneling, and threshold voltage effects, whereas Sharma et al. [30] proposed a subthreshold model addressing gate misalignment in double-gate architectures. Sai et al. [31] extended DC and RF modeling to cryogenic InGaAs MOSFETs, and Dastjerdy et al. [32] implemented three-dimensional non-equilibrium Green's function (NEGF) simulations for nanowire MOSFETs to elucidate quantum transport. Ghosh et al. [33] developed a drain current model incorporating drain-induced barrier lowering for dual-material cylindrical and surround-gate MOSFETs, demonstrating enhanced drain current, reduced output conductance, and superior performance over single-gate counterparts.

At the device physics level, Takagi et al. [34] and Lee [35] investigated the universal mobility–electric field relationship, while Jeon and Burk [36] and Chen et al. [37] formulated physically based semi-empirical models for drain current and inversion charge density. Li et al. [38] proposed an overcurrent protection integrated circuit (IC) to enhance SiC MOSFET reliability, and Wang et al. [39] experimentally and numerically investigated heavy-ion irradiation effects on *p*- and *n*-channel power UMOSFETs, analyzing leakage current degradation and single-event failure mechanisms such as burnout and gate rupture.

Despite substantial progress in both empirical and physics-based modeling, achieving a unified and accurate description of charge carrier mobility in silicon MOSFETs remains a critical challenge. Conventional single-model formulations often struggle to account for the complex interplay among temperature, doping concentration, electric field, and various scattering mechanisms. As device geometries continue to shrink, these limitations become increasingly pronounced, leading to deviations in simulated performance compared to experimental observations. Furthermore, balancing physical accuracy with computational efficiency remains a persistent concern, particularly in large-scale circuit simulations where model stability and predictive reliability are essential. These challenges underscore the need for refined mobility models capable of integrating multiple empirical formulations to enhance both precision and consistency across operating conditions.

The primary aim of this study is to develop and evaluate combined empirical modeling approaches to

improve the accuracy of charge carrier mobility simulation in silicon MOSFETs. Three hybrid models, Power Law–Lombardi, Arora–Lombardi–Caughey-Thomas, and Arora–Fletcher–Lombardi–Caughey-Thomas, are formulated and assessed against a baseline power-law model. The objective is to identify an optimal hybrid configuration that provides a robust representation of temperature, doping concentration, electric field, and scattering effects while maintaining computational efficiency. Validation against experimental data ensures that the proposed hybrid models deliver reliable predictive capability, offering a pathway toward more precise and efficient MOSFET performance simulation.

2 Mathematical modeling

2.1 Device structure and parameters

A silicon-based MOSFET is an electrical device composed of multiple material layers designed to regulate the flow of electric current. The basic structure of an n-channel MOSFET begins with a p-type doped silicon substrate. On this substrate, two heavily doped n^+ regions are formed to create the source and drain, which serve as the injection and collection terminals for charge carriers, respectively. The channel region lies between the source and drain, and its conductivity is controlled by the gate voltage. A thin silicon dioxide (SiO_2) layer acts as the gate oxide, providing electrical insulation between the gate and the channel, while a conductive polysilicon electrode forms the gate itself. When a voltage is applied to the gate, an electric field is established, attracting charge carriers to the substrate surface and creating a conductive path between the source and drain. Finally, metal contacts are deposited on the source, drain, and gate to enable external electrical connections. Owing to its simple yet highly efficient design, the MOSFET functions as both a switch and an amplifier in modern electronic circuits.

In this study, the MOSFET model employs a simplified 2D geometry with an overall device size of approximately 3 µm in width and 0.7 µm in height, representing a cross-sectional view as shown in Figure 1. The gate oxide interface maintains continuity of the electric displacement field. Trap-assisted recombination is represented using the Shockley–Read–Hall model, which describes the recombination of electrons and holes via defect states within the band gap, thereby enabling precise simulation of carrier behavior in the MOSFET. An overview of the parameters and silicon characteristics used in our simulation is provided in Table 1, with further details available in references [40–43].

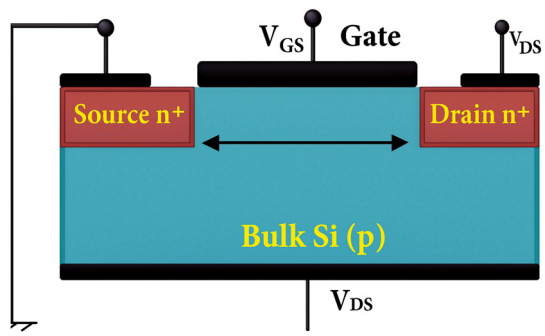

Figure 1: Abstract diagram of an N-channel transistor highlighting essential electrical connections.

Table 1: Key parameters and material properties of silicon.

Parameters	Value [40–43]
Silicon properties	
Relative permittivity (ε_{rnSi})	11.5
Band gap (E_g, eV)	1.12
Intrinsic carrier concentration (n_i, 1/cm^3)	1.5×10^{10}
Electron affinity (χ, eV)	4.05
Source/drain doping (N_D, 1/cm^3)	10^{20}
Substrat doping (N_A, 1/cm^3)	10^{17}
Oxide properties	
Relative permittivity (ε_{rnox})	4.5
Thickness (d, nm)	30

2.2 Carrier transport model and governing equations

The Boltzmann Transport Equation (BTE) serves as the theoretical foundation for understanding transport phenomena, describing the evolution of the charge carrier distribution function under disequilibrium conditions [44, 45]. In practice, device simulations frequently use the drift-diffusion approximation because of the intricacy of the Boltzmann equation [46, 47]. This approach, based on the assumption of local carrier equilibrium, represents charge transport as the combined effects of drift, driven by an electric field, and diffusion, resulting from concentration gradients [48, 49]. Models derived from this approximation significantly reduce computational time while maintaining the flexibility to study non-equilibrium transport in MOSFETs. However, empirical mobility models are necessary to accurately capture the influence of electric fields, and a more precise formulation is required for describing high-field transport and carrier diffusion [50, 51].

Here, the main charge transport equations in the drift–diffusion approximation are presented, namely the

continuity and Poisson equations. Electron and hole continuity equations are expressed as [48]:

$$\frac{\partial n}{\partial t} = G_n + \frac{1}{q}\nabla \cdot j_n, \tag{1}$$

$$\frac{\partial p}{\partial t} = G_p + \frac{1}{q}\nabla \cdot j_p, \tag{2}$$

where G_n and G_p represent the generation and recombination rates of electron–hole pairs. The drift–diffusion current densities for electrons and holes are given as [46, 52]:

$$\vec{j}_n = qn\mu_n\vec{E} + qD_n\vec{\nabla}n, \tag{3}$$

$$\vec{j}_p = qp\mu_p\vec{E} - qD_p\vec{\nabla}p. \tag{4}$$

Finally, the electric field (E) is determined using Poisson's equation [46]:

$$\nabla^2\phi = -\nabla \cdot E = q\left(p - n + N_a - N_d\right). \tag{5}$$

Equations (1)–(4) are coupled with Poisson's equation (5). It is important to note that the carrier mobilities μ_n and μ_p are not constant; they depend on temperature, electric field, doping concentration, and other physical factors.

2.3 Effective mobility

The MOSFET operates by modulating the conductivity of a semiconductor channel, typically silicon, through an electric field induced by the gate voltage. When the gate voltage exceeds a certain threshold, an inversion layer is formed at the semiconductor–oxide interface, enabling current to flow between the source and drain regions. The carrier mobility within this inversion layer is significantly lower than that in the bulk semiconductor and is referred to as the effective mobility (μ_{eff}). Charge transport in the channel is influenced by several scattering mechanisms, including phonon interactions, interface roughness at the gate oxide, Coulomb scattering from charged impurities, and structural defects, which collectively degrade mobility and device performance.

Three primary scattering mechanisms can be identified that limit carrier mobility. Each mechanism is influenced differently by factors such as temperature, transverse electric field, doping concentration, and other material or device parameters [34, 36, 53].

- Coulomb interactions, with charged impurities, limit mobility at low field.
- Interactions with phonons, for a temperature lower than 100 K; this type of collision is due to vibrations of the crystal lattice [54].
- Mobility is limited by collisions caused by surface roughness (channel/gate dielectric interface), particularly at high fields, because the carriers are trapped close to the interface.

Usually, mobility is described, as a first approximation, by Matthiessen's law (valid at low source–drain electric fields). This law assumes independent scattering mechanisms, allowing the probabilities of collisions with Coulomb centers (*cb*), surface roughness (*sr*), and phonons (*ph*) to be summed [48]:

$$\tau^{-1} = \sum_i \tau_i^{-1}, \tag{6}$$

that is,

$$\mu_{\text{eff}}^{-1} = \sum \mu_i^{-1}. \tag{7}$$

The relaxation time (τ) takes into account all the interactions that the carriers may undergo during transport, where i correspond to the relaxation time for a particular interaction [36]:

$$\frac{1}{\mu} = \frac{1}{\mu_{cb}} + \frac{1}{\mu_{sr}} + \frac{1}{\mu_{ph}}. \tag{8}$$

These parameters are determined using empirical models, which will be discussed in the following section, as the effects of small-scale physical phenomena are incorporated into the large-scale mobility used in transport equations.

2.4 Limitations of individual empirical mobility models

One of the most important parameters for describing charge carrier movement and current formulation in semiconductor devices is the electron and hole mobility, denoted as μ_n and μ_p, respectively. These mobilities have a significant impact on the electrical characteristics of MOSFETs. Various mobility models have been proposed in the literature to describe MOSFET behavior [55]. A few commonly used carrier mobility models are outlined below.

2.4.1 Power law model

Early studies of mobility introduced simple formulations such as the power law model [43]. This model, characterized by its simplicity, expresses the dependence of mobility on temperature through a power-law relationship [46]:

$$\mu_{nl} = \mu_{n,0} \times \left(\frac{T}{300}\right)^{-\alpha_n}, \tag{9}$$

$$\mu_{pl} = \mu_{p,0} \times \left(\frac{T}{300}\right)^{-\alpha_p}. \tag{10}$$

This model was commonly used in early device simulations when computational limitations restricted model

complexity. However, its accuracy is limited because it neglects the effects of doping concentration and electric field dependence.

2.4.2 Caughey–Thomas model

The 1960s saw the emergence of more sophisticated models, notably the Caughey–Thomas model. This model introduced the influence of the electric field on carrier mobility [56], simulating the saturation of carrier velocity under strong electric fields. The model is expressed as:

$$\mu_n(E) = \mu_{n0}\left[1 + \left(\frac{\mu_{n0}E_n}{v_{\text{sat},n}}\right)^{\beta_n}\right]^{-\frac{1}{\beta_n}}. \tag{11}$$

$$\mu_p(E) = \mu_{p0}\left[1 + \left(\frac{\mu_{p0}E_p}{v_{\text{sat},p}}\right)^{\beta_p}\right]^{-\frac{1}{\beta_p}}. \tag{12}$$

The default approach for calculating the saturation velocities incorporates temperature dependence and is defined as:

$$v_{\text{sat},n} = v_{n,0}\left(\frac{T}{T_{\text{ref}}}\right)^{\alpha_{2,n}}, \tag{13}$$

$$v_{\text{sat},p} = v_{p,0}\left(\frac{T}{T_{\text{ref}}}\right)^{\alpha_{2,p}}, \tag{14}$$

$$\beta_n = \beta_{n,0}\left(\frac{T}{T_{\text{ref}}}\right)^{\alpha_{1,n}}, \tag{15}$$

$$\beta_p = \beta_{p,0}\left(\frac{T}{T_{\text{ref}}}\right)^{\alpha_{1,p}}, \tag{16}$$

The default reference temperature is $T_{\text{ref}} = 300\,\text{K}$. The dimensionless material parameters, $\beta_{n,0}$, $\beta_{p,0}$, $\alpha_{1,n}$, $\alpha_{1,p}$, $\alpha_{2,n}$, and $\alpha_{2,p}$, are obtained from experimental data. By accounting for velocity saturation, this model became fundamental for simulating short-channel devices. However, it does not include the effects of doping concentration or detailed temperature dependence.

2.4.3 Arora et al.'s carrier mobility model

This model is temperature-dependent and includes both phonon and impurity scattering [57]. It is used to simulate the doping-dependent mobility:

$$\mu_n = \mu_{\min,n}\left(\frac{T}{300}\right)^{\beta_1} + \frac{\mu_{0,n}\left(\frac{T}{300}\right)^{\beta_2}}{1 + \left(\frac{N}{N_{0,n}} \cdot \left(\frac{300}{T}\right)^{\beta_3}\right)^{\alpha}}, \tag{17}$$

$$\mu_p = \mu_{\min,p}\left(\frac{T}{300}\right)^{\beta_1} + \frac{\mu_{0,p}\left(\frac{T}{300}\right)^{\beta_2}}{1 + \left(\frac{N}{N_{0,p}}\right)^{\alpha}}, \tag{18}$$

with:

$$N = N_a^- - N_d^+, \tag{19}$$

and

$$\alpha = \alpha_0\left(\frac{T}{300}\right)^{\beta_4}. \tag{20}$$

This model, while remaining relatively simple, has become a valuable tool for power device simulations in which doping plays a crucial role. However, it remains limited as it does not account for the dependence on the electric field.

2.4.4 Lombardi model

The late 1980s marked a turning point with the introduction of the Lombardi model [58], a semi-empirical formulation grounded in physical principles to describe the scattering mechanisms influencing carrier mobility. It accounts for phonon scattering, ionized impurity scattering, and interface roughness effects. The general expression applies Matthiessen's rule to combine these mechanisms as follows:

$$\frac{1}{\mu} = \frac{1}{\mu_0} + \frac{1}{\mu_{ac}} + \frac{1}{\mu_{sr}}, \tag{21}$$

where

$$\mu_{ac,n} = \frac{\mu_{1,n}}{\left(\frac{E_\perp}{E_{\text{ref}}}\right)} + \frac{\mu_{2,n}\left(\frac{N}{N_{\text{ref}}}\right)^{\alpha_n}}{\left(\frac{E_\perp}{E_{\text{ref}}}\right)^{\frac{1}{3}}\left(\frac{T}{T_{\text{ref}}}\right)}, \tag{22}$$

$$\mu_{sr,n} = \frac{\delta_n}{E_\perp^2}, \tag{23}$$

$$\mu_{ac,p} = \frac{\mu_{1,p}}{\left(\frac{E_\perp}{E_{\text{ref}}}\right)} + \frac{\mu_{2,p}\left(\frac{N}{N_{\text{ref}}}\right)^{\alpha_p}}{\left(\frac{E_\perp}{E_{\text{ref}}}\right)^{\frac{1}{3}}\left(\frac{T}{T_{\text{ref}}}\right)}, \tag{24}$$

$$\mu_{sr,p} = \frac{\delta_p}{E_\perp^2}, \tag{25}$$

with

$$N = N_a^- + N_d^+. \tag{26}$$

This model provides a more accurate fit to experimental data and a deeper physical understanding of the underlying transport phenomena. However, it is more complex to implement than earlier empirical models and requires careful parameter calibration to achieve optimal accuracy.

2.4.5 Dorkel–Leturg model

This model is used to describe scattering effects caused by thermally excited acoustic phonons, doping impurities, and carrier–carrier interactions [59]. Carrier–carrier scattering mobility (μ_{ccs}) becomes particularly relevant under high carrier concentrations. The following equation defines this model [59]:

$$\mu_L = \mu_{L0} \times \left(\frac{T}{300}\right)^{-\alpha}, \tag{27}$$

which accounts for lattice or phonon scattering effects on electron and hole mobility, also known as lattice mobility (μ_L). Here, μ_{L0} and α depend on the dopant type and temperature range, while T represents the lattice temperature. When both carrier types are present in high densities, carrier–carrier scattering mobility (μ_{ccs}) becomes significant, and the Fletcher mobility model can be applied [60]:

$$\mu_{ccs} = \frac{F_1\left(\frac{T}{300}\right)^{\frac{3}{2}}}{\sqrt{pn}\ln\left(1+\frac{F_2\left(\frac{T}{300}\right)^2}{(pn)^{-\frac{1}{3}}}\right)}, \tag{28}$$

to describe the effect of carrier–carrier interactions on overall mobility.

To account for interactions between charge carriers and ionized impurities, impurity mobility (μ_I) is introduced [59]:

$$\mu_I = \frac{AT^{\frac{3}{2}}}{N}\left[\ln\left(1+\frac{BT^2}{N}\right)-\frac{BT^2}{N+BT^2}\right]^{-1}. \tag{29}$$

where N denotes the doping concentration, and A and B are parameters dependent on the dopant type.

These models provide analytical expressions capturing mobility reduction due to enhanced phonon interactions at elevated temperatures, increased ionized impurity scattering at higher doping levels, and carrier–carrier interactions under high injection conditions. They are particularly useful for simulating devices where high injection effects are important. However, the models have limitations: they rely on simplified representations of scattering mechanisms, exhibit limited accuracy under high electric fields, assume a relatively homogeneous semiconductor, and depend on parameters fitted to experimental data. As such, they are primarily designed for silicon devices and are less accurate for high-field transport modeling.

2.5 Combining empirical mobility models

Charge transport in semiconductors involves multiple physical mechanisms, which cannot be fully captured by a single mobility model. Combining several models enables a more comprehensive description of how carrier mobility is influenced by temperature, doping concentration, electric fields, and scattering processes. In this work, we adopt a hybrid modeling approach, evaluating multiple combinations of empirical models to generate different scenarios and assess their effectiveness in accurately predicting MOSFET behavior.

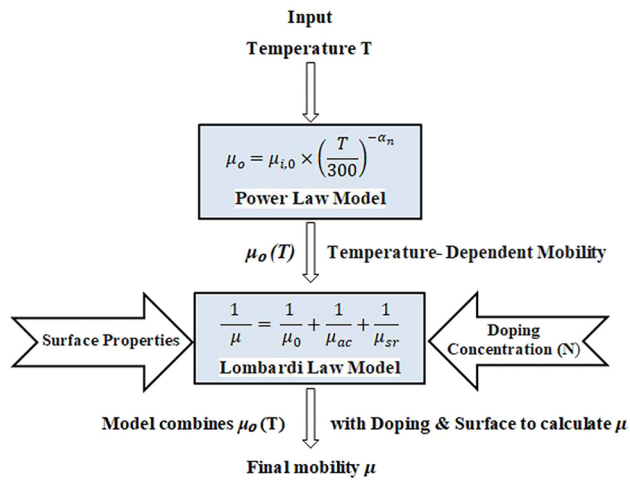

Figure 2: Model integrating various scattering processes.

2.5.1 Scenario (i): combined power Law–Lombardi model

The first scenario employs a hybrid approach that combines the Power Law and Lombardi models. The Power Law model calculates a baseline low-field mobility that depends on temperature, which is then used as a key input for the more physically detailed Lombardi model as shown in Figure 2. The Lombardi model refines this baseline by accounting for the effects of phonon scattering, ionized impurity scattering, and surface roughness scattering. This combination provides a computationally efficient and reasonably accurate representation of carrier mobility, capturing both temperature dependence and multiple scattering mechanisms. The resulting mobility is subsequently used in the carrier transport equations for device simulation.

2.5.2 Scenario (ii): combined Arora, Lombardi, and Caughey-Thomas models

The combination of the Arora, Lombardi, and Caughey-Thomas mobility models provides a comprehensive approach for modeling charge transport in MOSFET structures, capturing a wide range of physical effects as illustrated in Figure 3. The Arora model establishes a foundational description of mobility, accounting for temperature and doping concentration dependence, which is essential for accurate simulation across different operating conditions and device doping profiles. According to this, the Lombardi model enhances the mobility

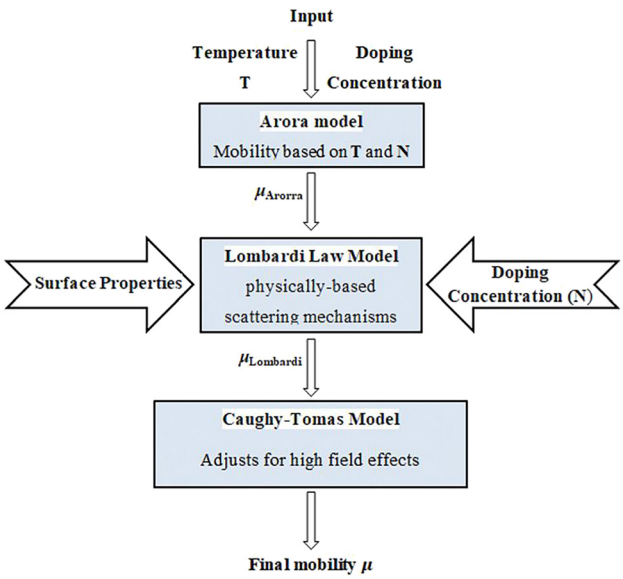

Figure 3: Model incorporating temperature, doping, and high-field effects.

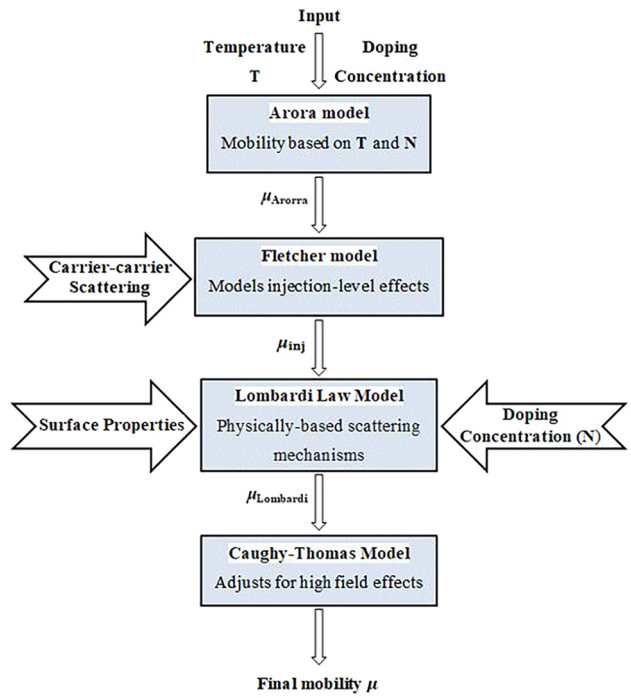

Figure 4: Model combination, by capturing the interplay of temperature, doping, injection level, scattering mechanisms, and electric field effects.

computation by adding the physically-based mechanisms of scattering, i.e., phonon, ionized impurity, and surface roughness scattering, and thus improves the reliability of simulations within the channel area. The Caughey-Thomas model addresses the limitations of the previous two models under high electric field conditions by modeling velocity saturation, which is particularly important in short-channel MOSFETs near the drain region.

In this combined approach, the Arora model first provides a baseline mobility value, which is then successively adjusted by the Lombardi model and the Caughey-Thomas model. This cascading methodology captures both high-field and temperature-dependent effects, resulting in a more accurate representation of carrier mobility. The integration of these models enhances the realism of simulation results, providing valuable insights for MOSFET design, analysis, and optimization.

2.5.3 Scenario (iii): combined Arora, Fletcher, Lombardi, and Caughey-Thomas models

To simulate carrier transport accurately across a wide range of operating conditions, a combined mobility modeling approach is employed, integrating the strengths of multiple established models. Building on Fletcher's work on high-current effects [60], this scenario incorporates injection-level effects into the mobility computation. In this approach, the baseline mobility calculated by the Arora model is first modified by the Fletcher model to account for reductions caused by increased carrier–carrier interactions, and is then further refined by the Lombardi and Caughey-Thomas models, as illustrated in Figure 4. This cascading methodology captures the combined influence of temperature, doping, scattering mechanisms, high electric fields, and injection-level effects, resulting in a comprehensive and realistic representation of carrier transport for semiconductor device simulations.

2.6 Empirical model parameterization and calibration

The accuracy of empirical mobility models strongly depends on the proper calibration of numerous adjustable parameters, including minimum electron mobility, minimum hole mobility, low-field electron and hole velocities, and temperature or field exponents. Specifically, parameters such as $\mu_{\min,p}$, $\mu_{\min,n}$, $\upsilon_{n,0}$, $\upsilon_{p,0}$, $\beta_{n,0}$, $\alpha_{1,n}$ and others play a critical role in defining the behavior of charge carriers under different operating conditions.

The primary objective of this study is to evaluate the structural effectiveness of different model combinations across varying operational regimes, rather than performing device-specific optimization. Consequently, widely accepted standard parameter sets for bulk silicon, derived from extensive experimental fitting data reported in the literature, are employed.

For Scenarios i, ii, and iii, all model coefficients are obtained from the original studies by Caughey-Thomas [56], Arora [57], and Lombardi [58], and correspond to values commonly used in generalized silicon devices. These parameters ensure that the base mobilities, field dependencies, and temperature exponents accurately represent the semi-classical transport behavior of silicon, providing a robust basis for assessing the performance of the combined mobility models.

3 Numerical modeling

3.1 Numerical simulation

The numerical simulation of a two-dimensional Si MOSFET was conducted following a structured workflow to ensure accuracy, reproducibility, and computational efficiency, as outlined in Figure 5. The procedure comprises three sequential phases: Pre-Processing, Solving, and Post-Processing.

In the Pre-Processing phase, the finite element type, device geometry, material properties, and mobility model parameters were defined, followed by the construction of a two-dimensional device representation and the generation of a computational mesh with local refinement in critical regions. Boundary conditions were imposed, and output variables were specified.

During the solving phase, static or dynamic analysis was selected, and the global matrices and element vectors were assembled over the domain. The coupled nonlinear partial differential equations, encompassing Poisson's equation for the electrostatic potential and the carrier continuity equations for electrons and holes, were solved iteratively, with convergence criteria rigorously monitored to ensure numerical stability and accuracy.

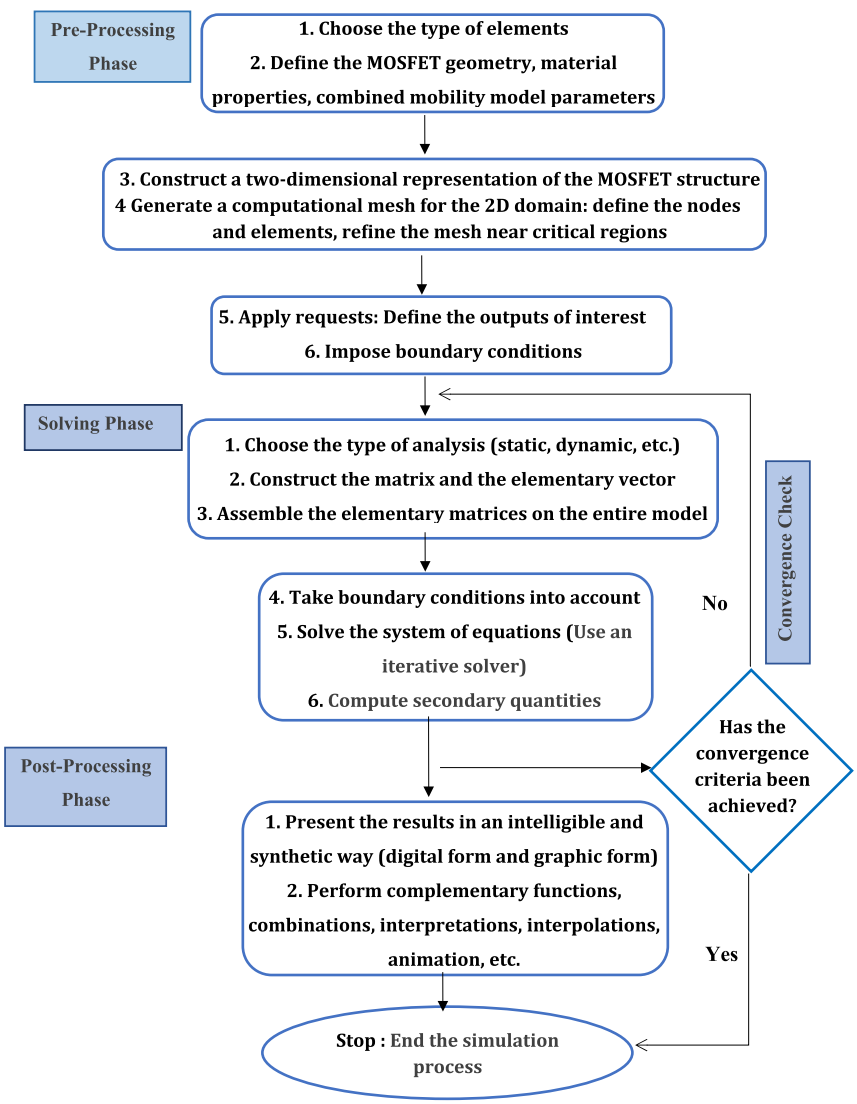

Figure 5: Analysis process using a numerical model.

In the post-processing phase, converged solutions were analyzed both quantitatively and graphically, with additional operations, including interpolation, combination, and derived quantity evaluation, performed to extract relevant physical insights and validate device performance. This methodology ensures that the numerical model reliably captures the electrical characteristics of the MOSFET while minimizing computational errors.

3.2 Numerical validation

Figure 6 illustrates the variation of the calculated effective mobility as a function of the transverse electric field at three distinct lattice temperatures: 133 K (low-temperature regime where phonon scattering is minimal), 297 K (near room temperature), and 447 K (high-temperature regime characterized by dominant phonon scattering). The simulated results (dashed lines) are compared with the experimental data (symbols) reported by Takagi et al. [34] for a substrate doping concentration of 3.9×10^{15} cm^{-3}. The strong agreement between the simulated and experimental data across the examined temperature range validates the accuracy of the developed numerical model. The model accurately reproduces the expected mobility enhancement at 133 K due to the suppression of lattice vibrations and the corresponding reduction in phonon-limited scattering, as well as the significant mobility degradation at 447 K resulting from increased phonon–carrier interactions. This consistency confirms that the implemented combined mobility model provides a reliable physical representation of carrier transport mechanisms under varying thermal conditions.

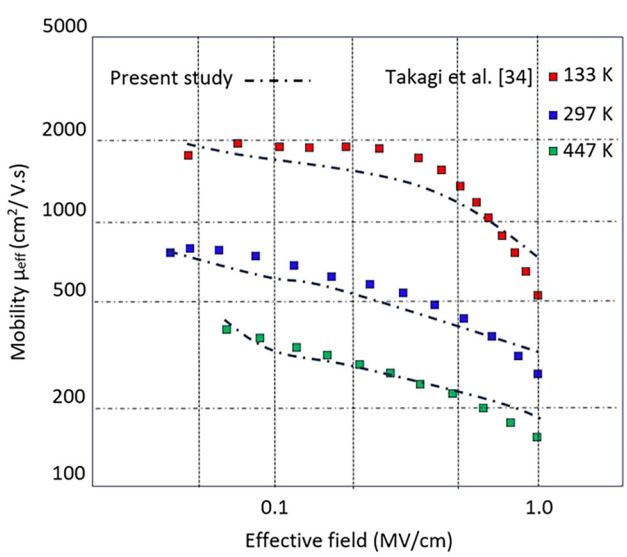

Figure 6: Simulated and experimental effective mobility versus transverse electric field at different lattice temperatures for a substrate doping concentration of 3.9×10^{15} cm^{-3}.

4 Results and discussions

Figure 7 presents the drain current (I_d) versus gate voltage (V_g) characteristics for a MOSFET operating at a low drain voltage of $V_d = 10$ mV, simulated under three different modeling scenarios in addition to the constant mobility case. The curves depict the transistor's response across both the subthreshold and strong inversion regions. In Figure 7(a), the constant mobility model yields an almost linear behavior but fails to accurately represent the subthreshold conduction and the saturation effects observed in strong inversion. The second configuration (Scenario i), which combines the Power Law model (to account for temperature dependence) and the Lombardi model (to capture surface scattering effects), produces a smoother transition between the subthreshold and strong inversion regions, though minor deviations in leakage current estimation are observed. The third configuration (Scenario ii), which further incorporates the Arora model (doping-dependent mobility) and the Caughey–Thomas model (mobility saturation at high electric fields), improves accuracy by yielding a steeper subthreshold slope and a clearer transition. Finally, the fourth configuration (Scenario iii), which additionally integrates the Fletcher model (for temperature dependence at low temperatures), provides the most comprehensive representation, exhibiting enhanced leakage current prediction and a more pronounced transition between operating regions.

Figure 7(b), which provides a magnified view around the threshold voltage, shows that the constant mobility model produces a diffuse transition between the subthreshold and strong inversion regions. The first two combined models improve this transition; however, it remains less pronounced than in the third and fourth combined models, where the inclusion of doping- and temperature-dependent effects sharpens the transition and yields a more accurate threshold voltage (V_{th}).

Figure 7(c), which presents I_d on a logarithmic scale, enables clearer visualization of subthreshold currents at low gate voltages and accentuates the distinctions among the mobility models. The influence of the various modeling approaches is evident: while the constant mobility model tends to overestimate or underestimate the leakage current, the third and fourth combined models provide more realistic predictions, particularly under low-temperature conditions, due to their inclusion of both temperature and doping effects. The estimated V_{th} in these models is approximately 1.15 V, closely matching the theoretical value of about

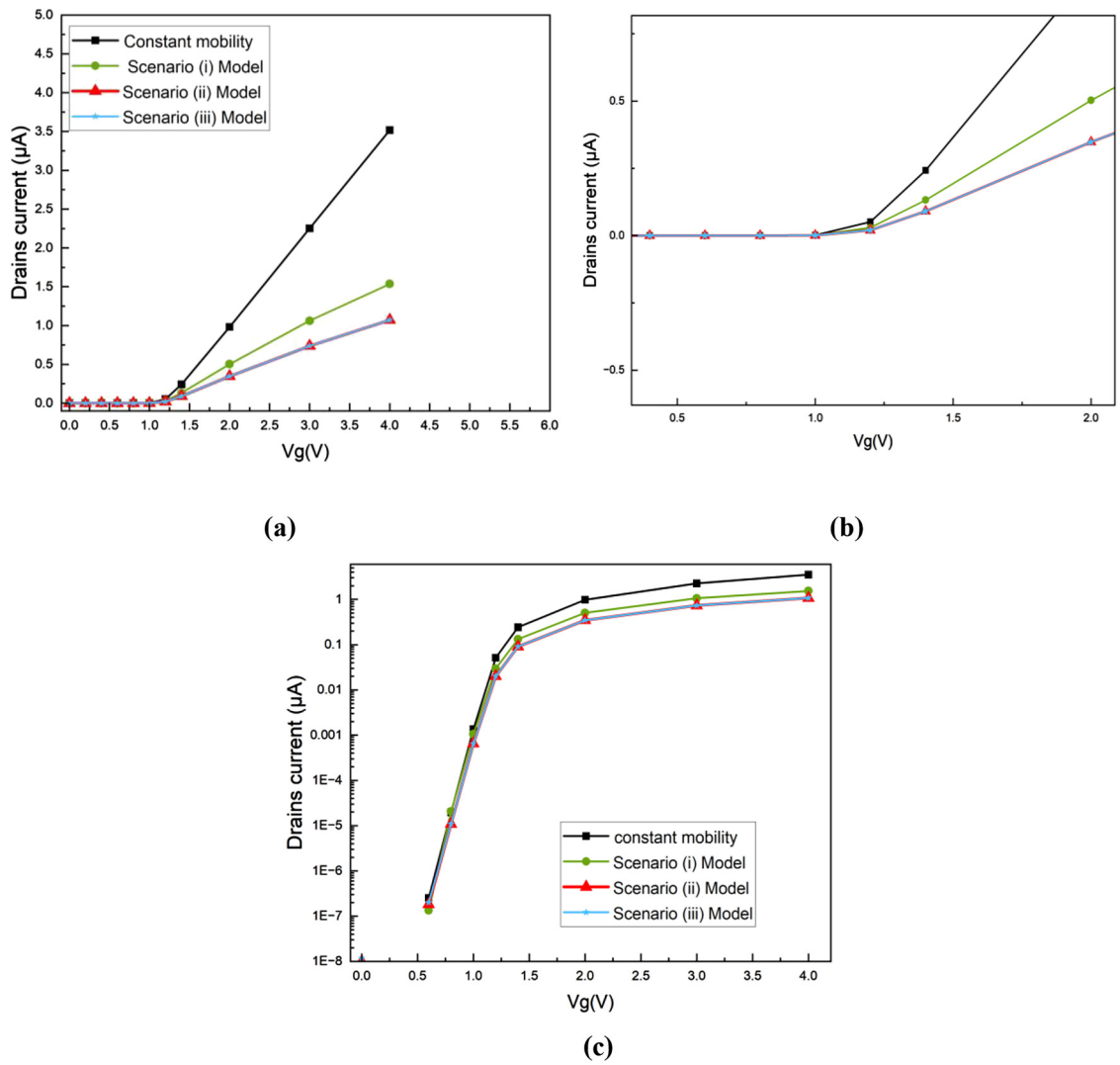

Figure 7: Comparison of I_d versus V_g characteristics across different combined mobility models: (a) linear-scale plot at low drain bias ($V_d = 10$ mV), (b) zoom-in view around the threshold voltage (V_{th}), and (c) logarithmic-scale representation highlighting subthreshold behavior.

1.27 V [34, 35]. Moreover, the leakage current (I_{off}) is more accurately captured by the third and fourth combined models, ranging between 10^{-12} A and 10^{-13} A at low temperatures, whereas the constant mobility and second combined models either overpredict or underpredict this parameter.

Figure 8 displays I_d as V_d varies with different gate voltages: $V_g = 2$ V in Figure 8(a), 3 V in Figure 8(b), and 4 V in Figure 8(c), simulated under three different scenarios as well as with constant mobility. These curves illustrate the MOSFET's behavior in both the linear (triode) and saturation regimes, allowing for a visual comparison of the I_d–V_d characteristics obtained with different models.

Several key differences can be observed: with constant mobility, the I_d–V_d curve in the saturation region exhibits a nonzero slope, which is unrealistic. In Scenario (i), the I_d–V_d curve flattens further in the saturation region but may still not be perfectly horizontal. In Scenarios (ii) and (iii), the I_d–V_d curve is closer to a horizontal line in the saturation region, aligning more closely with the real-world behavior of a MOSFET. The combined model curves exhibit a shallower slope compared to the constant mobility curve, indicating that mobility is reduced due to the incorporation of mobility models. This figure also highlights the impact of different mobility models on current behavior in the linear region. In the saturation region, the combined mobility models more accurately capture charge carrier velocity saturation, resulting in a more pronounced flattening of the I_d–V_d curve. Velocity saturation occurs when charge carriers reach a maximum velocity beyond which they can

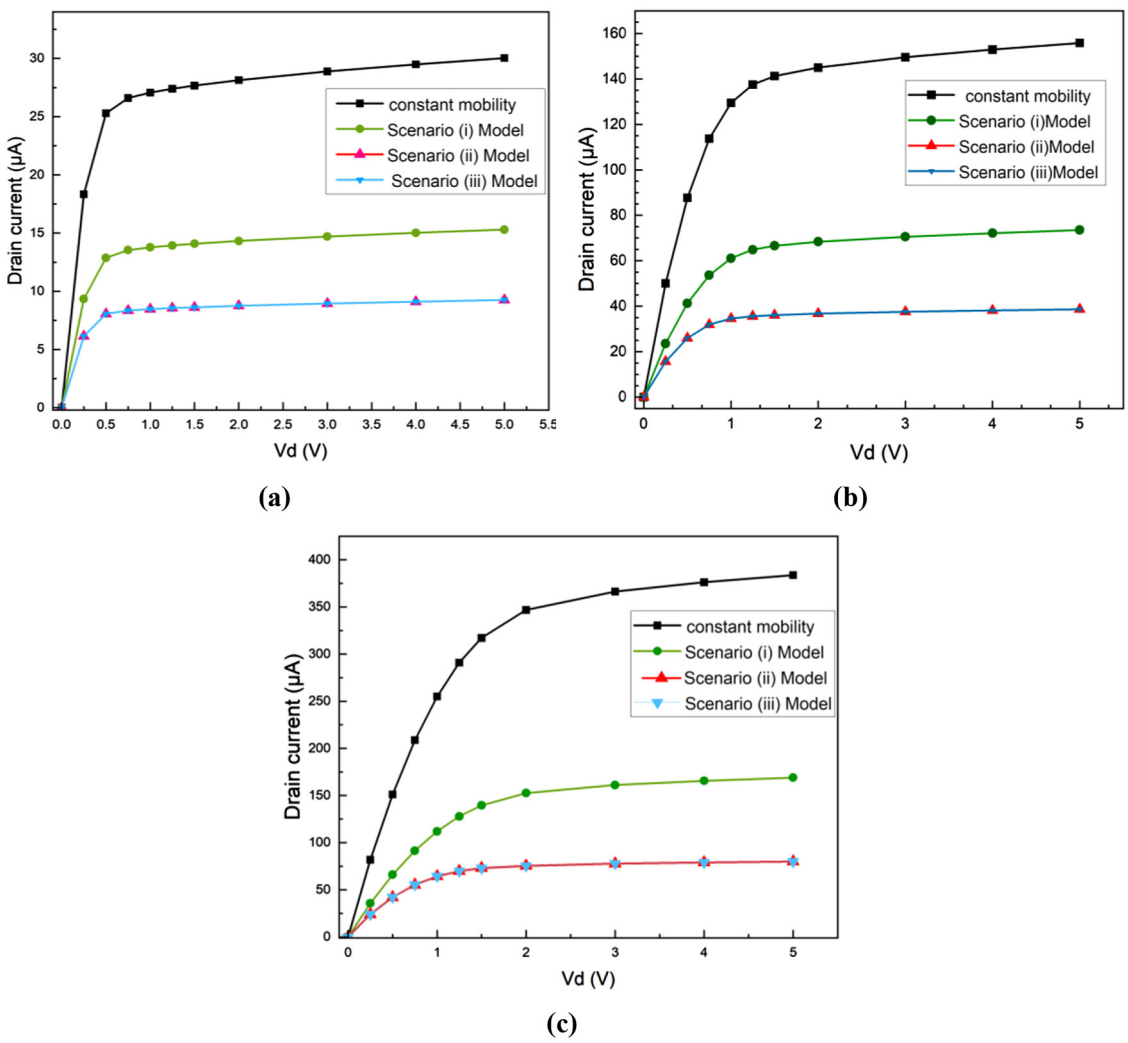

Figure 8: I_d versus V_d characteristics for various mobility models, illustrating performance in the linear and saturation regimes: (a) $V_g = 2$ V, (b) $V_g = 3$ V, (c) $V_g = 4$ V.

no longer accelerate, even as the electric field continues to increase.

Figure 9(a) shows that the peak electric field near the drain side of the MOSFET increases with increasing drain voltage. As depicted in Figure 9(b), the peak magnitude remains approximately constant across all cases. Figure 9(c) illustrates that the MOSFET structure has been optimized to minimize band-to-band tunneling (BTBT) leakage current, a critical factor in reducing power consumption and enhancing the overall performance and reliability of MOSFET-based circuits.

Figure 10 is the log plot of electron concentration versus position in the channel for the three cases. Relative to V_g, as V_g increases beyond the threshold voltage (V_{th}), the electron concentration in the channel also increases. In terms of channel position, the concentration is larger near the source and decreases gradually towards the drain, especially at high V_d. Near the drain, greater V_d reduces the concentration of carriers, eventually leading to pinch-off.

Figure 11 illustrates the electric potential distribution across the channel for various situations, with curves (a–c) showing the potential variation as a function of the V_g under a 1 V drain bias. It is evident that the potential drops in the inversion region as the V_g increases. Curves (d–f) demonstrate that increasing the V_d raises the potential barrier in the drain region. As the drain-to-source voltage

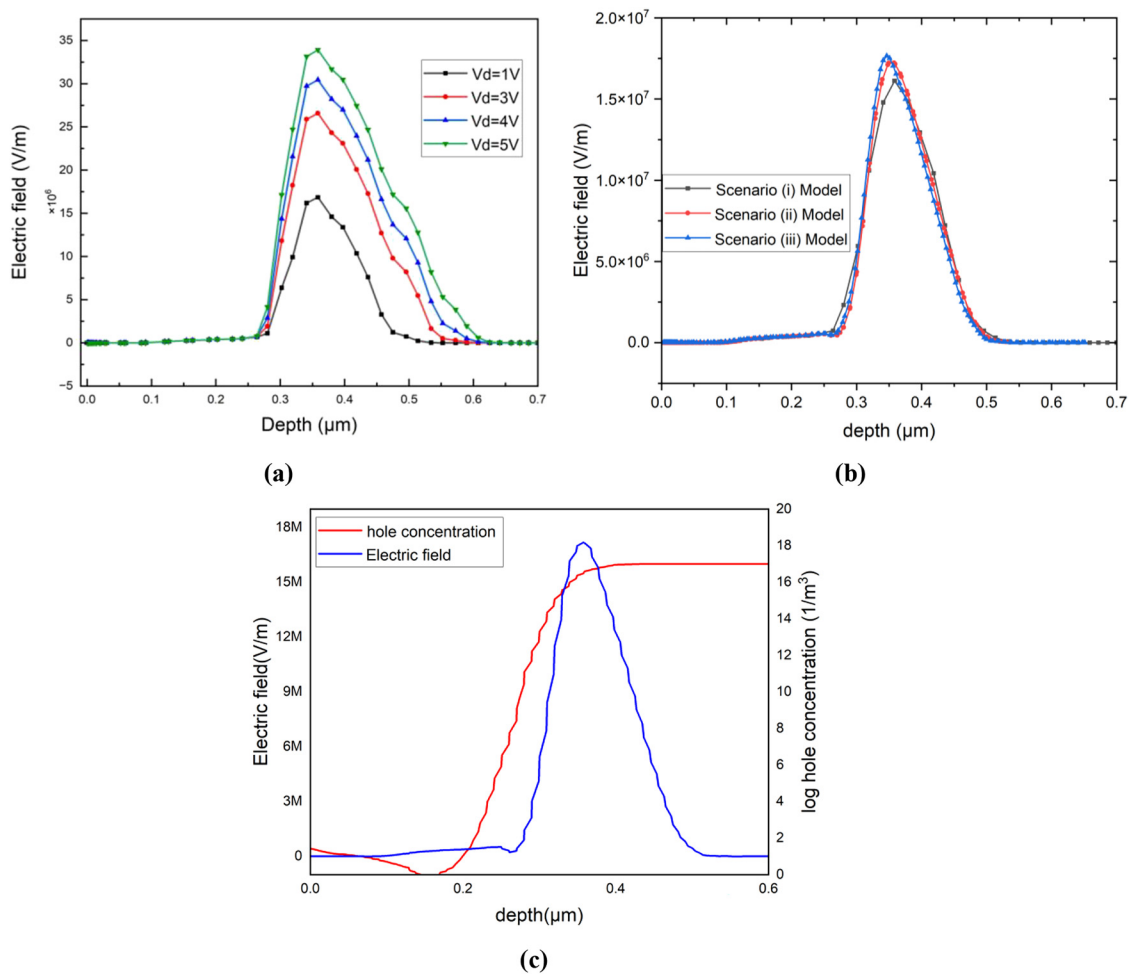

Figure 9: Analysis of the electric field in the MOSFET drain region: (a) peak electric field depth as a function of V_d at $V_g = 4$ V, (b) comparison of the electric field depth across the integrated mobility scenarios (i, ii, and iii) at $V_g = 4$ V, (c) electric field and hole concentration profile, demonstrating the optimization for BTBT minimization.

(V_{ds}) rises, an electric field is established along the channel, which accelerates charge carriers from the source toward the drain. This electric field is stronger near the drain than near the source. When V_{ds} reaches a value such that $V_{gs} - V_{ds} = V_{th}$, the concentration of charge carriers near the drain becomes very low or even zero, causing the channel to 'pinch off' near the drain. Although the channel is pinched, current continues to flow due to the electric field between the drain and the pinched region.

In the saturation region of the MOSFET, charge carriers are progressively drawn from the channel toward the drain. Under these conditions, the I_d becomes nearly independent of the V_{ds} and primarily depends on the V_{gs}. Figure 12 compares the electron concentration and electric potential distributions along the channel at $V_g = 4$ V and $V_d = 4$ V for the three scenarios, assuming constant mobilities. It is important to note that the mobility models significantly influence the redistribution of carriers under the applied voltage, thereby affecting the carrier concentration profile. With constant mobility, the pinch-off occurs abruptly, and the electron concentration drops sharply to zero near the drain region. This approach often yields poor estimates near the drain, as it neglects the effects of doping concentration and high electric fields. In Scenario (i), the carrier concentration is slightly higher than in the constant mobility case, although the near-drain variations remain somewhat underestimated. This model improves accuracy by incorporating temperature and surface scattering effects but omits doping and field-dependent mobility influences. In Scenarios (ii) and (iii), where field-dependent mobility

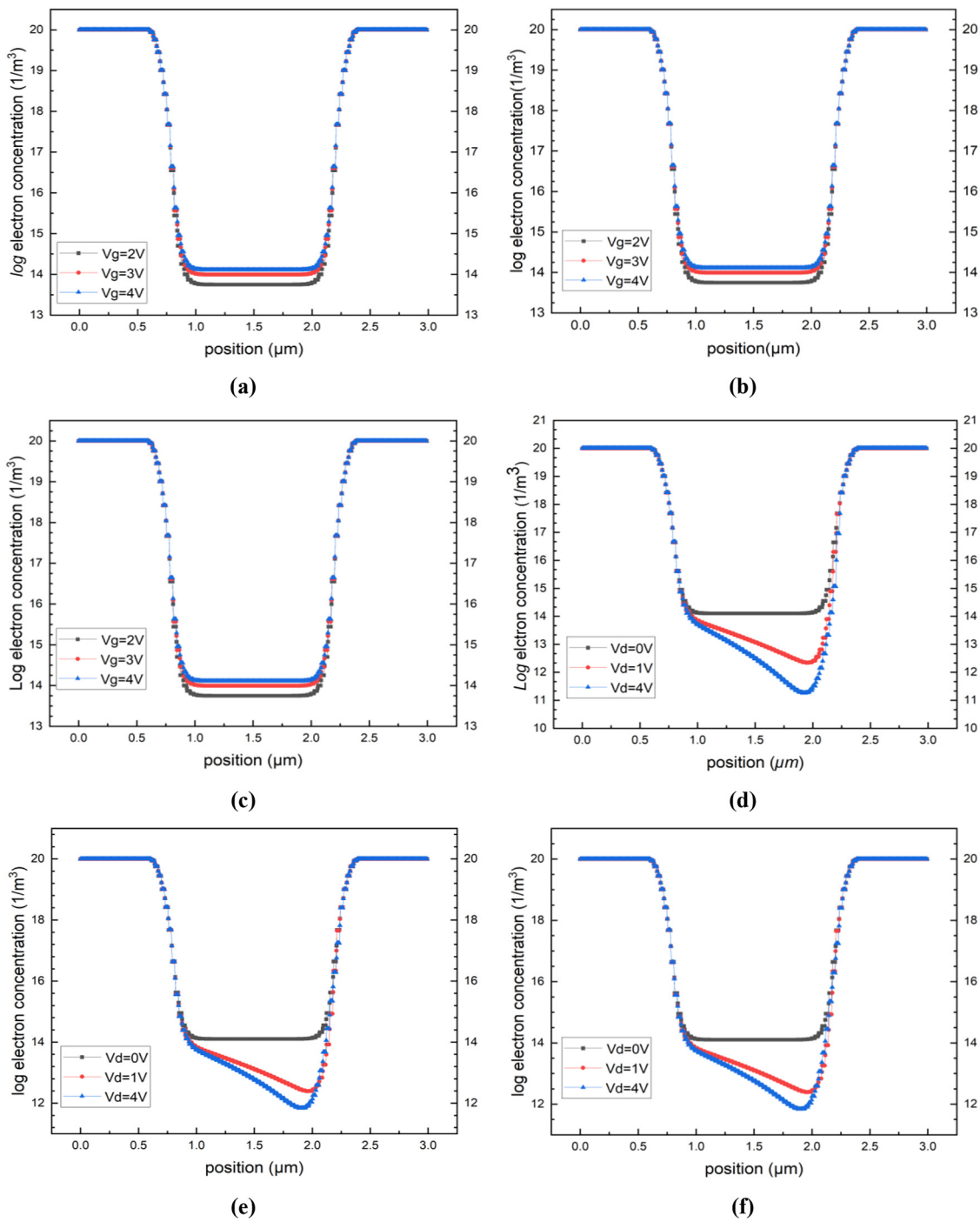

Figure 10: Logarithm of electron concentration profile along the MOSFET channel: (a)–(c) inversion channel formation and density evolution at V_d for increasing V_g, (d)–(f) impact of increasing V_d under a high gate bias ($V_g = 4$ V).

models are employed, the pinch-off behavior becomes more gradual, with the carrier concentration decreasing progressively toward the drain, providing a more realistic representation of the physical process.

These models account for velocity saturation effects that occur under strong electric fields. As the electric field intensity increases, carrier mobility decreases, thereby limiting the rise in current. This behavior directly affects both

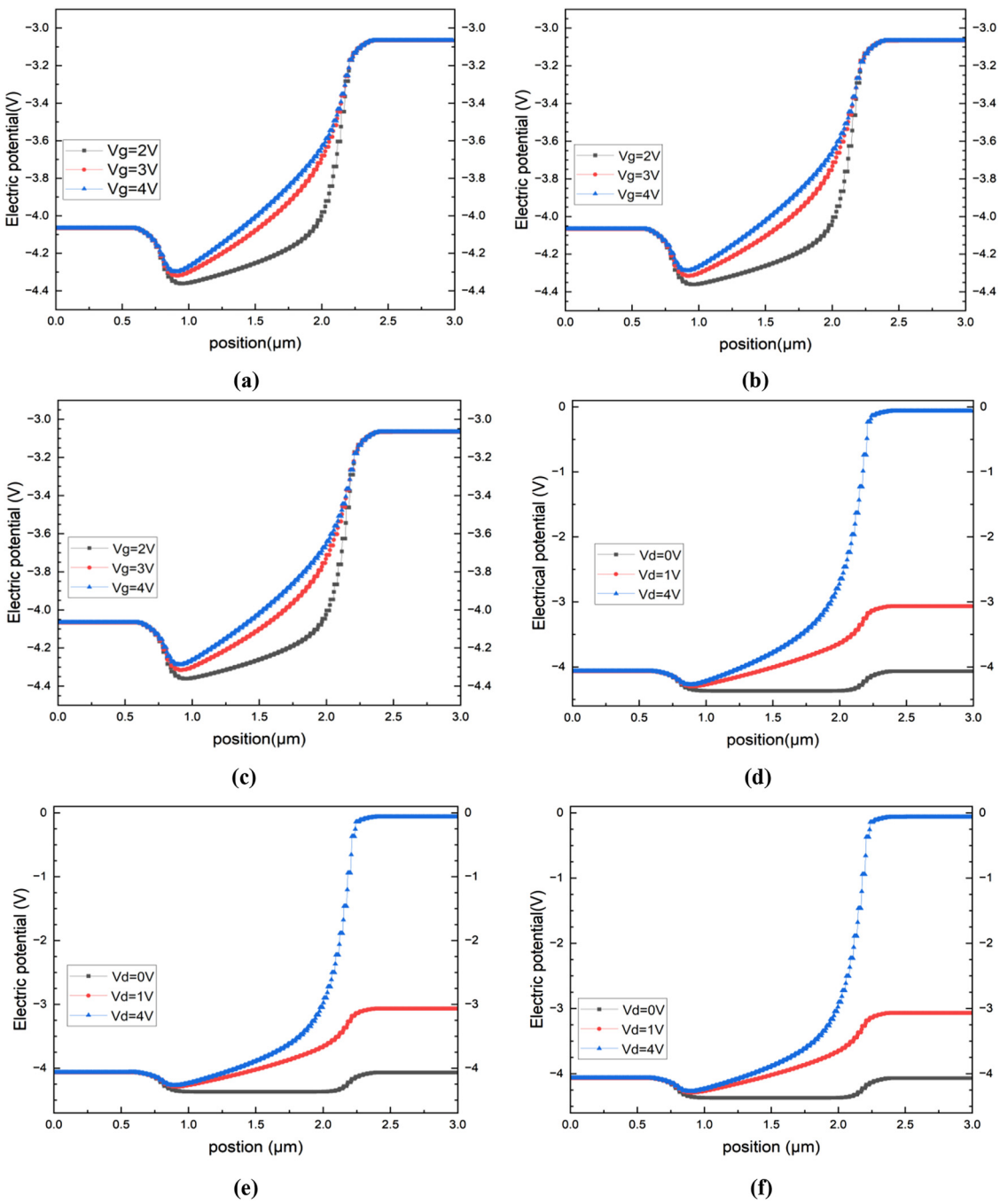

Figure 11: Electric potential distribution across the MOSFET channel: (a)–(c) evolution of the inversion layer potential with increasing V_g at a fixed drain bias ($V_d = 1$ V), (d–f) effect of increasing V_d under a high gate bias ($V_g = 4$ V).

the potential and charge carrier distributions. In the pinch-off region near the drain, where the electric field is highest, the reduction in mobility leads to a higher carrier concentration compared to that predicted by the constant mobility model. Consequently, the pinch-off effect appears less abrupt in the field-dependent mobility models, resulting in a smoother and more physically realistic transition of carrier concentration along the channel.

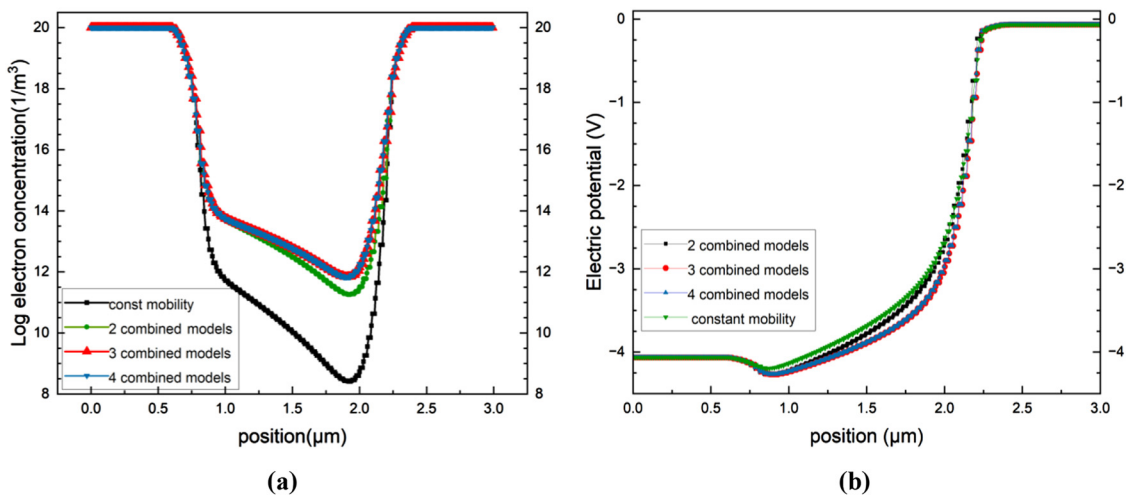

Figure 12: Impact of mobility modeling on carrier distribution and electrostatic behavior in the saturation regime ($V_g = 4$ V, $V_d = 4$ V): (a) electron concentration profile and (b) electric potential distribution along the channel.

5 Conclusions

This study conducted a comprehensive evaluation of various empirical mobility models to simulate silicon MOSFET behavior, emphasizing both accuracy and computational efficiency. The principal contribution lies in the systematic comparative assessment of different model integration strategies, establishing a structured framework for optimizing the trade-off between precision and computational cost in semiconductor device simulation.

Among the investigated configurations, Scenario (iii), which integrates the Fletcher model with the Arora, Lombardi, and Caughey–Thomas models to account for high-level injection effects, produced results closely matching those of Scenario (ii). Detailed analysis revealed that the relatively low injection levels under the examined operating conditions minimized the influence of the Fletcher model, rendering its additional complexity unnecessary. Consequently, Scenario (ii), which combines the Arora, Lombardi, and Caughey–Thomas models, was identified as the optimal configuration, providing accurate and reliable predictions of MOSFET behavior while maintaining superior computational efficiency.

The integration of these empirical mobility models substantially enhances the physical realism of carrier transport representation by simultaneously accounting for temperature dependence, doping concentration, and high-field mobility saturation, effects often neglected in single-model approaches. The results confirm that Scenario (ii) delivers the best balance between numerical accuracy and computational feasibility, offering a validated and practical guideline for model selection in both academic and industrial semiconductor simulation platforms.

Despite these advancements, the study acknowledges inherent limitations of classical empirical formulations. Their predictive accuracy for ultra-scaled devices (below the 20 nm technology node), where quantum confinement and tunneling effects become dominant, remains uncertain. Accordingly, future research will focus on developing hybrid quantum–empirical models that combine the validated Scenario (ii) framework with quantum correction techniques such as density-gradient theory. This hybridization will enable efficient yet physically consistent modeling of nanoscale MOSFETs, capturing critical quantum phenomena that influence inversion charge distribution, threshold voltage behavior, and subthreshold conduction.

Overall, the combined empirical mobility models proposed herein represent a significant advancement in the accurate and computationally efficient simulation of conventional and moderately scaled MOSFETs. More importantly, they establish a clear foundation for extending this methodology toward next-generation semiconductor technologies, where coupling classical transport models with quantum-mechanical corrections will be essential for predictive, high-fidelity device simulation.

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Data availability statement: The datasets generated and/or analysed during the current study are available from the corresponding author on reasonable request.

Nomenclature

A, B	parameters depending on the nature of the dopants
D_n, D_p	electron and hole diffusion coefficients (m^2/s)
E	electric field (V/m)
$E_\perp$	perpendicular electric field component (V/m)
F_1	fletcher mobility coefficient (s^2 A/(m^3 kg))
F_2	fletcher mobility coefficient ($1/m^2$)
G_n	rate of generation and recombination of electron-hole pairs (cm^{-3} s^{-1})
j_n, j_p	electron and hole current densities (A/m^2)
μ_0	electron and hole mobility reference (cm^2/V s)
μ_1, μ_2	model parameters for electrons and holes (cm^2/V s)
μ_{ccs}	carrier-carrier scattering mobility (cm^2/V s)
μ_{eff}	effective mobility (cm^2/V s)
μ_I	impurity mobility (cm^2/V s)
μ_L	lattice mobility (cm^2/V s)
μ_n, μ_p	electron and hole mobilities (cm^2/V s)
$\mu_{\min,n}$	electron mobility minimum reference (cm^2/V s)
$\mu_{\min,p}$	hole mobility minimum reference (cm^2/V s)
$\mu_{n,0}$, $\mu_{p,0}$	low-field electron and hole mobilities (cm^2/V s)
$\mu_{0,n}$, $\mu_{0,p}$	electron and hole mobility reference (cm^2/V s)
n, p	electron and hole concentrations (cm^{-3})
N	doping concentration (cm^{-3})
$N_{0,n}$	electron reference impurity concentration (cm^{-3})
$N_{0,p}$	hole reference impurity concentration (cm^{-3})
N_a^-	ionized acceptor concentration (cm^{-3})
N_d^+	ionized donor concentration (cm^{-3})
ϕ	electrostatic potential (V)
ρ	charge density (C/m^3)
τ	relaxation time (s)
T	lattice temperature (K)
T_{ref}	reference temperature (K)
V_d	drain voltage (V)
V_g	gate voltage (V)
V_{th}	threshold voltage (V)
$\upsilon_{n,0}$	saturation velocity for electrons (cm/s)
$\upsilon_{p,0}$	saturation velocity for holes (cm/s)
α_n, α_p	material properties
$\alpha_{1,n}$, $\alpha_{1,p}$	dimensionless model parameters
$\alpha_{2,n}$, $\alpha_{2,p}$	dimensionless model parameters
$\beta_{n,0}$, $\beta_{p,0}$	dimensionless model parameters
δ	model parameter for electrons and holes

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